

BaseO3CPU 参数修改 — 对齐 OpenC910

修改文件: `src/cpu/o3/BaseO3CPU.py`

修改汇总

参数	原值	新值	C910 RTL 依据
fetchWidth	8	3	ct_ifu_top.v:382-386
decodeWidth	8	3	ct_idu_top.v:824-828
renameWidth	8	4	ct_idu_ir_dp.v:123-155
dispatchWidth	8	4	ct_rtu_rob.v:31-46
wbWidth	8	3	ct_rtu_pst_preg.v:6132-6134
commitWidth	8	3	ct_rtu_rob.v:449-451
LQEntries	32	16	ct_lsu_lq.v:159
SQEntries	32	12	ct_lsu_sq_entry.v:614
numPhysIntRegs	256	95	ct_idu_rf_prf_pregfile.v
numPhysFloatRegs	256	32	ct_idu_rf_prf_eregfile.v
numPhysVecRegs	256	64	ct_idu_rf_prf_vregfile.v
numROBEntries	192	64	ct_rtu_rob.v

Diff

```
@@ -83,7 +83,7 @@ class BaseO3CPU(BaseCPU):
    renameToFetchDelay = Param.Cycles(1, "Rename to fetch delay")
    iewToFetchDelay = Param.Cycles(1, "Issue/Execute/Writeback to fetch
delay")
    commitToFetchDelay = Param.Cycles(1, "Commit to fetch delay")
-   fetchWidth = Param.Unsigned(8, "Fetch width")
+   fetchWidth = Param.Unsigned(3, "Fetch width")
    fetchBufferSize = Param.Unsigned(64, "Fetch buffer size in bytes")
    fetchQueueSize = Param.Unsigned(
        32, "Fetch queue size in micro-ops per-thread"

@@ -98,14 +98,14 @@ class BaseO3CPU(BaseCPU):
    # Forward pipeline delays
    bacToFetchDelay = Param.Cycles(1, "Branch address calc. to fetch
delay")
    fetchToDecodeDelay = Param.Cycles(1, "Fetch to decode delay")
-   decodeWidth = Param.Unsigned(8, "Decode width")
+   decodeWidth = Param.Unsigned(3, "Decode width")
```

```

        iewToRenameDelay = Param.Cycles(
            1, "Issue/Execute/Writeback to rename delay"
        )
        commitToRenameDelay = Param.Cycles(1, "Commit to rename delay")
        decodeToRenameDelay = Param.Cycles(1, "Decode to rename delay")
-       renameWidth = Param.Unsigned(8, "Rename width")
+       renameWidth = Param.Unsigned(4, "Rename width")

@@ -116,15 +116,15 @@ class BaseO3CPU(BaseCPU):
    issueToExecuteDelay = Param.Cycles(
        1, "Issue to execute delay (internal to the IEW stage)"
    )
-   dispatchWidth = Param.Unsigned(8, "Dispatch width")
+   dispatchWidth = Param.Unsigned(4, "Dispatch width")
    issueWidth = Param.Unsigned(8, "Issue width")
-   wbWidth = Param.Unsigned(8, "Writeback width")
+   wbWidth = Param.Unsigned(3, "Writeback width")

    iewToCommitDelay = Param.Cycles(
        1, "Issue/Execute/Writeback to commit delay"
    )
    renameToROBDelay = Param.Cycles(1, "Rename to reorder buffer delay")
-   commitWidth = Param.Unsigned(8, "Commit width")
+   commitWidth = Param.Unsigned(3, "Commit width")

@@ -139,8 +139,8 @@ class BaseO3CPU(BaseCPU):
    5, "Time buffer size for forward communication"
    )

-   LQEntries = Param.Unsigned(32, "Number of load queue entries")
-   SQEntries = Param.Unsigned(32, "Number of store queue entries")
+   LQEntries = Param.Unsigned(16, "Number of load queue entries")
+   SQEntries = Param.Unsigned(12, "Number of store queue entries")
    LSQDepCheckShift = Param.Unsigned(
        4, "Number of places to shift addr before check"
    )

@@ -172,12 +172,12 @@ class BaseO3CPU(BaseCPU):
    numRobs = Param.Unsigned(1, "Number of Reorder Buffers")

    numPhysIntRegs = Param.Unsigned(
-       256, "Number of physical integer registers"
+       95, "Number of physical integer registers"
    )
    numPhysFloatRegs = Param.Unsigned(
-       256, "Number of physical floating point registers"
+       32, "Number of physical floating point registers"
    )
-   numPhysVecRegs = Param.Unsigned(256, "Number of physical vector registers")
+   numPhysVecRegs = Param.Unsigned(64, "Number of physical vector registers")
    numPhysVecPredRegs = Param.Unsigned(
        32, "Number of physical predicate registers"
    )

```

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)  
  
@@ -185,7 +185,7 @@ class BaseO3CPU(BaseCPU):  
    # most ISAs don't use condition-code regs, so default is 0  
    numPhysCCRegs = Param.Unsigned(0, "Number of physical cc registers")  
    instQueues = VectorParam.IQUnit(IQUnit(), "Vector of IQs")  
-    numROBEntries = Param.Unsigned(192, "Number of reorder buffer  
entries")  
+    numROBEntries = Param.Unsigned(64, "Number of reorder buffer entries")
```

参数分类

流水线宽度 (6 项)

- **fetchWidth** 8 → 3: 取指宽度, 受 IFU 每周期取指能力限制
- **decodeWidth** 8 → 3: 译码宽度, 与取指匹配
- **renameWidth** 8 → 4: 重命名宽度, 略大于译码以支持多发射
- **dispatchWidth** 8 → 4: 分发宽度, 与重命名匹配
- **wbWidth** 8 → 3: 写回宽度, 与 commit 匹配
- **commitWidth** 8 → 3: 提交宽度, ROB 每周期可提交指令数

队列大小 (3 项)

- **LQEntries** 32 → 16: Load Queue 条目数
- **SQEntries** 32 → 12: Store Queue 条目数
- **numROBEntries** 192 → 64: ROB 总条目数

物理寄存器 (3 项)

- **numPhysIntRegs** 256 → 95: 整型物理寄存器数
- **numPhysFloatRegs** 256 → 32: 浮点物理寄存器数
- **numPhysVecRegs** 256 → 64: 向量物理寄存器数

保持不变

- **issueWidth** 8: C910 issueWidth 为 8, 与原值一致
- **squashWidth**: RTL 中不存在此概念, 保持未设置
- **numPhysVecPredRegs** / **numPhysMatRegs** / **numPhysCCRegs**: RTL 中不存在, 保持原值